Adarsh Dubey

Machine Learning Engineer | Applied AI Scientist | ML Systems and Kernel Engineer

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Professional Summary

Applied machine learning engineer and data scientist with an M.Tech in Machine Learning and Computing (IIST) and an M.Sc. in Mathematics (GATE 2024 All India Rank 245). End-to-end experience building and evaluating deep learning and large language model (LLM) systems: transformer training and inference, generative AI and agentic AI pipelines, computer vision, and statistical experimentation with reproducible metrics. Industry work on GPU and NPU kernel optimization, distributed model training (DeepSpeed, FSDP), production LLM serving benchmarks (latency, throughput, cost, energy), and fintech document-AI workflows. Strong in Python, SQL, PyTorch, data analysis, model evaluation, and translating research ideas into measurable production trade-offs. Hackathon winner, IISc Workshop on Systems for Machine Learning 2026.

Technical Skills

- **Machine Learning and AI**: Machine Learning, Artificial Intelligence, Deep Learning, Statistical Modeling, Predictive Modeling, Supervised Learning, Neural Networks, Transformers, Large Language Models, Natural Language Processing, Computer Vision, Generative AI, Agentic AI, Reinforcement Learning, Model Training, Fine-tuning, Model Evaluation, Model Deployment, Hyperparameter Tuning, Feature Engineering, Experimentation, A/B Testing, Benchmarking, Hypothesis Testing, Probability and Statistics
- **Programming and Libraries**: Python, C++, CUDA, SYCL, Bash, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, FastAPI, Pydantic, Hugging Face ecosystem, vLLM, SGLang, DeepSpeed
- **Data and Databases**: SQL, SQLite, FTS5, relational modeling, Pandas, data analysis
- **ML Systems and Performance**: Distributed Training (DDP, FSDP, ZeRO), Federated Learning, LLM Inference, FlashAttention, KV Cache, GPU Programming, Kernel Optimization, CUTLASS, Intel Gaudi and XPU, TT-Metal, Energy Optimization, Latency Optimization, Profiling (perf, ftrace), Docker, Linux, Git, Redis, MQTT
- **Domain Experience**: Fintech, Banking Workflows, Market Microstructure, Time-Series Modeling, Edge AI, TinyML, Process Analytics, Document Verification

Professional Experience

•Neurealm Pvt. Ltd. *June 2026 - Present*

Engineer, ML Systems and Kernel Bangalore, India

- Designed and optimized **TT-Metal kernels** for high-throughput **matmul** and **FlashAttention** on the LLM prefill and decode path, using DRAM and L1 interleaved sharding plus padding to raise memory-bandwidth utilization, cut bank conflicts, and reduce tail latency in streaming token generation.
- Implemented kernel-level **normalization** (RMSNorm, softmax) and precision-aware datapaths with tiled L1 and DRAM layouts and lock-free buffers, resolving concurrency and memory-fragmentation defects while keeping **numerical stability** and deterministic **deep learning inference** performance.
- Working on **model porting** of transformer and vision-language models onto the **TI TDA54** automotive SoC (C7 NPU with heterogeneous CPU and DSP fabric): mapping tiled matmul and attention onto on-chip SRAM, and designing the scheduling path for **edge AI** and **agentic AI** multi-step inference under real-time power and latency constraints.
- Developed **sliding-window FlashAttention** with block classification (fully-inside, intersecting, fully-outside) to enable selective computation, cutting DRAM traffic and delivering measurable compute and memory savings in narrow-context regimes.

•Intel Technology India Pvt. Ltd. *June 2025 - May 2026*

AI Framework Engineer (Intern) Bangalore, India

- Designed a **Sliding Window Flash Attention (SWFA)** sparse-attention algorithm with block classification, enabling selective computation and **65% compute savings** for narrow-window configurations; the proof-of-concept with per-position masking delivered **50% training speedup** at sub-microsecond precision across **19 test configurations**.
- Designed and ran a large-scale **LLM inference benchmark** of 7B to 70B models on **Intel Gaudi2 versus Gaudi3**: **7,199** prompt-level measurements over **36** device configurations logging time-to-first-token, inter-token latency, throughput, and power. Gaudi3 led throughput by up to **38%**; Gaudi2 stayed **1.9x** better on energy per token, guiding production deployment and cost decisions.
- Implemented **distributed training** with **DeepSpeed ZeRO** and **FSDP** for scalable LLM training at **93.8% GPU utilization** across multi-GPU configurations, and validated **TorchFT** fault recovery on Intel XPU for production AI workloads.
- Contributed **FusedQKNormRope** and **FusedQKRope** kernels to Intel's public `sglang_xpu_kernel` repository, optimizing XPU **LLM inference** performance for SGLang serving.
- Optimized compile-time performance for **CUTLASS Xe_gemm** operations, maintained AI framework compatibility across Intel Xe series GPUs, and presented a CUTE and CUTLASS tensor-layout, HBM-to-SLM memory deep-dive to the CUTLASS infrastructure team.

Projects and Research

•High-Performance Deep Learning Software Stack for Scalable and Efficient LLMs *July 2025 - May 2026*

M.Tech Thesis, IIST

- Built a high-performance deep learning stack for scalable **LLM training** using distributed strategies (DeepSpeed ZeRO, FSDP, DDP) and a novel sliding-window block-sparse attention algorithm; derived the masking, proved numerical equivalence to dense attention, and measured **93.8% GPU utilization**, **65% compute savings**, and **40% energy-efficiency improvement**. *Stack: PyTorch Distributed, DeepSpeed, vLLM, CUDA kernels.*

•**LLM Inference Benchmarking: Intel Gaudi2 versus Gaudi3** *2025 - 2026*

Applied ML and ML Systems | Code

– Designed the experiment and evaluated dense, mixture-of-experts, and vision-language models across two accelerator generations with vLLM and Optimum Habana, deriving energy per token from power traces. Inter-token latency fell **19%** on Gaudi3 while Gaudi2 delivered **2.27 versus 4.35 J/token**: the throughput-optimal and energy-optimal hardware are not the same. *Stack: Python, vLLM, Optimum Habana, Pandas, Matplotlib.*

•**Centralized to Federated Edge Learning on a Raspberry Pi Cluster (Hackathon Winner)** *Feb 2026*

Workshop on Systems for Machine Learning 2026, IISc Bengaluru | Code

– Took AlexNet, VGG16, and ResNet101 from a 2x Tesla T4 centralized baseline to a live **4-node federated learning** cluster on Raspberry Pi with non-IID splits (Flotilla, MQTT, Redis, Docker). **FedProx 77.3%** versus FedAvg 71.8% over 10 rounds; selected AlexNet for deployment on training time and a **13.5%** cross-dataset generalization gap rather than accuracy alone. *Stack: PyTorch, Flower-style FL, MQTT, Redis, Docker.*

•**IASW: Agentic AI Document Verification for Core Banking** *2026*

Generative AI and Fintech | Code

– Built a four-agent pipeline (validation, OCR and vision-language field extraction, confidence scoring, summary) automating the maker role for core-banking account changes on a local **Qwen2.5-VL-7B** with zero external API calls; the database write path is reachable only through the human-checker endpoint, enforcing human-in-the-loop in code. *Stack: Python, FastAPI, Pydantic, SQLite, OCR, VLM.*

•**KYC Policy Graph RAG** *2026*

Generative AI and Knowledge Graphs | Code

– Built a production-shaped **Graph RAG** for KYC/AML policy with PDF ingestion, validated schema/LLM extraction, and RRF fusion across MiniLM chunks, 2-hop graph edges, and Louvain communities. Added SQLite provenance, cited FastAPI responses, Docker/CI-ready delivery, and equal-k evaluation: graph **6/6** vs chunk RAG 5/6, MRR **1.00** vs 0.861; verified `ACTS_AS` → `CANNOT_WRITE` path and 26-triple extraction set. *Stack: Python, NetworkX, SQLite/FTS5, MiniLM, FastAPI, Docker, pytest.*

•**GPU-Native Limit Order Book Simulator with Statistical Replay** *2026*

Quantitative ML and Data Science | Code

– CUDA matching engine simulating thousands of parallel market scenarios under Hawkes-driven order flow; compared execution strategies with **paired statistical tests** and 95% confidence intervals (TWAP beat market orders by 24.90 bps implementation shortfall across 60 seeds). *Stack: CUDA, C++, Python, NumPy, SciPy.*

•**Guardian Edge: Instant Hazard Alert for Two-Wheelers** *Dec 2025 - Jan 2026*

ACM Winter School on Edge AI Hackathon, IISc | Code

– Real-time hazard detection and rider alerting on Arduino Nicla Vision using dual-IMU sensing and an **89-parameter on-device neural network**, delivering low-latency, energy-efficient **TinyML** inference at the edge. *Stack: Python, MicroPython, PyTorch, NumPy, Decision Tree, Neural Network.*

Education

•**Indian Institute of Space Science and Technology, Thiruvananthapuram** *2024 - 2026*

Master of Technology, Machine Learning and Computing

•**Jawaharlal Nehru University, New Delhi** *2022 - 2023*

Post Graduate Diploma, Big Data Analytics

•**Chhatrapati Shahu Ji Maharaj University, Kanpur** *2019 - 2021*

Master of Science, Mathematics

•**Chhatrapati Shahu Ji Maharaj University, Kanpur** *2016 - 2019*

Bachelor of Science, Computer Applications and Mathematics

Key Courses Taken

- **AI, ML, and Systems**: Foundations of Machine Learning, Advanced Machine Learning, Graphical Models and Deep Learning, Reinforcement Learning, Statistical Modelling, Computer Vision, Applied Accelerated AI, Introduction to Large Language Models, GPU Architectures and Programming, Optimization Techniques, Numerical Linear Algebra
- **Data and Methods**: Data Modeling, Big Data Analytics, Advanced Computational Methods for Optimization, Algorithms and Practical Implementation in Bioinformatics

Certifications and Achievements

•**Selected presenter**, INSPIRE 2026, IIST symposium for postgraduate research; presented M.Tech thesis on scalable LLM training *2026*

•**Hackathon Winner**, Workshop on Systems for Machine Learning, CDS, IISc Bengaluru *2026*

•**GATE 2024, Mathematics**, All India Rank 245 (top 0.2 percent) *2024*

•**Claude Certified Architect: Foundations (CCAF)**, Anthropic, generative AI systems design *2026*

•Workshops: Systems for LLMs (CSE, IIT Bombay); Hands-on LLMs (IIST); Walmart CTE Reinforcement Learning (CSA, IISc); ACM Winter School on Edge AI (IISc) *2025-2026*